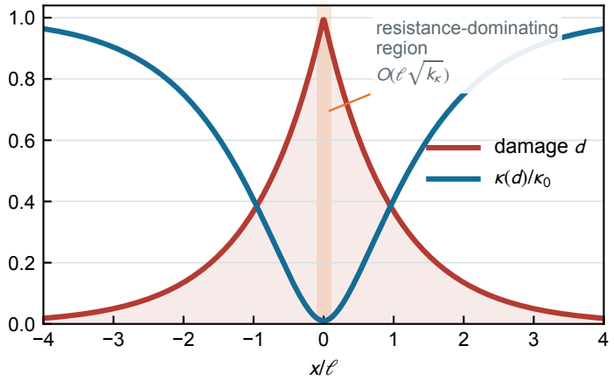
Diffuse cracks encode a finite thermal resistance

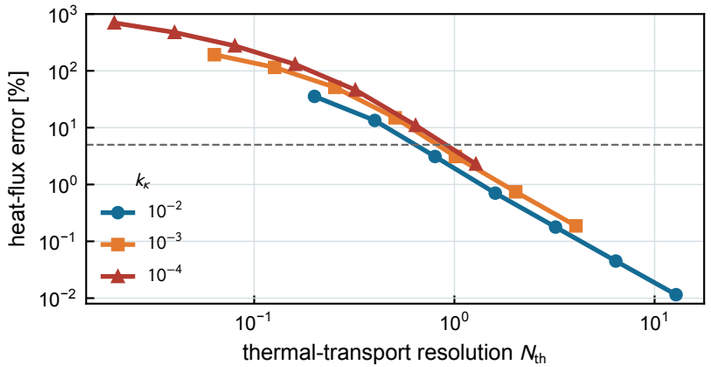
EXACT RESISTANCE | TRANSPORT RESOLUTION | THERMAL-SHOCK CONSEQUENCE

1 EXACT RESISTANCE



$$R_{\mathrm{pf}} = \frac{\ell}{\kappa_0} I(k_\kappa)$$

2 RESISTANCE CALIBRATION · FLUX CONVERGENCE



phase-field resolution ≠ transport convergence

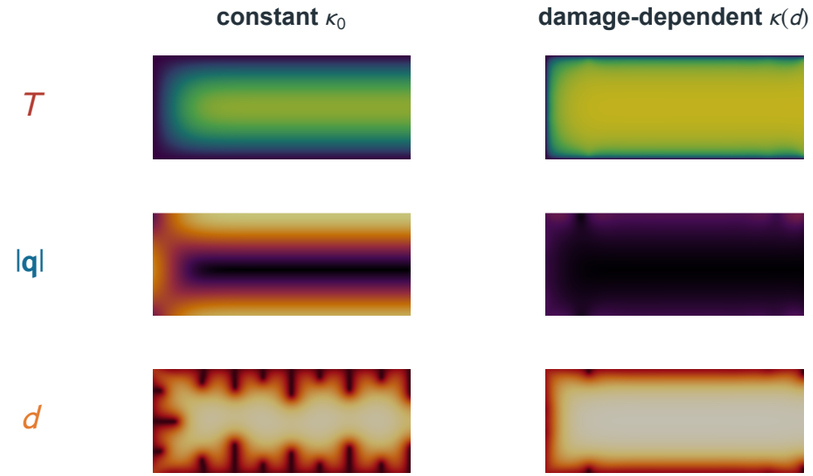
$$I(k_\kappa) = \kappa_0 R_* / \ell$$

resistance-equivalent calibration

$$N_{\mathrm{th}} = \ell\sqrt{k_\kappa}/h$$

transport-resolution criterion

3 THERMAL-SHOCK CONSEQUENCE



higher resistance → lower |q| → heat retention → weaker fracture drive